NAREN MUTHURAJ LATHA

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Project portfolio & Contact links

Hybrid Mechanical-Design Engineer combining a strong foundation in mechanical kinematics, fabrication, and mechatronics with an MSc in Design & Engineering from Politecnico di Milano. Proven ability to bridge theoretical structural analysis with physical prototyping, hardware-in-the-loop data acquisition, and closed-loop control systems, seeking a high-tech research internship at the University of Twente to contribute to multidisciplinary engineering research.

EDUCATION

Politecnico Di Milano, Italy

MSc in Design & Engineering

2025- Exp. October 2027

- Current Average: **27.66 / 30.0**
- **Focus Areas:** New Product Development, Mechanical Design, Design & Manufacturing, and Frameworks for Systematic Innovation.

Amrita Vishwa Vidyapeetham, India

B.Tech in Mechanical Engineering

2020-2024

- CGPA: 7.83 / 10.0
- **Bachelor's Thesis:** FE Simulation of Chip Formation in Oblique Cutting of Ti6Al4V (Numerical modeling in Abaqus/Explicit validating shear forces and material plastic deformation).

TECHNICAL SKILLS & METHODOLOGIES

- **Engineering CAD & Simulation:** SolidWorks, PTC Creo, Abaqus/Explicit (FEA), Autodesk Inventor, AutoCAD, Fusion360.
- **Design for Manufacturing (DFM):** Injection Molding (draft angles, wall thickness, parting lines), Sheet Metal Cold-Forming.
- **Prototyping & Fabrication:** CNC Drilling, Manual Metal Bending, TIG Welding, Woodworking, 3D Printing, Custom Jig Fabrication.
- **Mechatronics & Data Acquisition:** Arduino, I2C Sensor Integration, Automotive LIN Bus Architecture, Closed-Loop Control Systems.
- **Industrial Design & UI/UX:** Figma, Blender, CorelDRAW, Human-Centered Design, Applied Semiotics.

LANGUAGES

- English - Professional
- Italian - Basic

ACADEMIC RESEARCH & LAB PROJECTS

MO.EV.ES. Electric Mountain Cart | Design Studio II

- Conducted extensive user research to conceptualize an off-road, last-mile delivery electric cart designed for navigating alpine refugee campsites and rough terrain.
- Calculated 48V battery capacity and torque requirements for a 200kg payload on 25° inclines, integrating a commercial reduction gearbox and Motenergy 1117 motor.
- Replicated industrial fabrication processes by constructing custom bending jigs to manually form the aluminum and steel tubular chassis prototype.

Nebu 90° Medical Inhaler | Design Studio I

- Redesigned a portable nebuliser focusing on ergonomics and discreet aesthetics, utilizing rapid 3D printing to empirically validate user grip dimensions.
- Applied foundational DFM principles for ABS injection molding, maintaining a strict 2mm uniform wall thickness across all shells to prevent sink marks.
- Conceptualized a proprietary magnetic pogo-pin umbilical acting as an electronic position-sensing interlock to prevent accidental user exposure to the UV self-cleaning cycle.

Frugal Design & Fabrication Workshop | Chalmers University of Technology

- Fabricated a 1:1 scale contaminant-resistant sheet metal food enclosure utilizing strictly handheld workshop machinery.
- Troubleshot material shearing limits during hydraulic pressing by modifying the blank-holder friction on custom-built wooden stamping dies.

INDEPENDENT ENGINEERING & MECHATRONICS

Closed-Loop Electromechanical Exhaust Controller | System Architecture

- Fabricated a switchable automotive exhaust bypass system, upcycling OEM steel piping via TIG welding to withstand extreme thermal cycling.
- Architected the control logic (PWM torque-boosts, debounce algorithms) for an Arduino/L298 H-bridge system, utilizing AI tools to orchestrate and generate the code.
- Integrated an ACS712 current sensor to detect the mechanical stall spike at the valve's end-of-travel, instantly cutting motor power to prevent burnout.

Automotive LIN Bus Architecture Retrofit | Protocol Feasibility Study

- Conducted an architectural study to bridge an MQB-platform steering wheel onto a legacy PQ25 vehicle network.
- Mapped the theoretical integration of dual TJA1021 LIN transceivers to sniff, translate, and inject cruise control frames across mismatched topologies.

PROFESSIONAL EXPERIENCE & PUBLICATIONS

Schneider Electric, India

2023

Electro-Mechanical Design Intern

- Engineered a space-constrained kinematic door interlock for a draw-out Electrically Operated Module (EOM) using PTC Creo.
- Designed a compliant L-bracket mechanism that actuates along the breaker's force vector, safely restricting high-voltage access during service conditions.

Published Research

2024

Measurement Science and Technology

- Co-authored published research on machine learning-based Tool Condition Monitoring (TCM) systems for predictive maintenance.
- Engineered the physical Data Acquisition (DAQ) architecture, securely mounting MPU6050 accelerometers to VMC spindles and configuring Arduino I2C logging to capture tool vibration data for the analytics team.